Elmo's Cookie Waves

Euler's Equation Makes Waves, and What Phase Means

November 4, 2026 · Nathaniel

Name: _____ Date: _____

The story

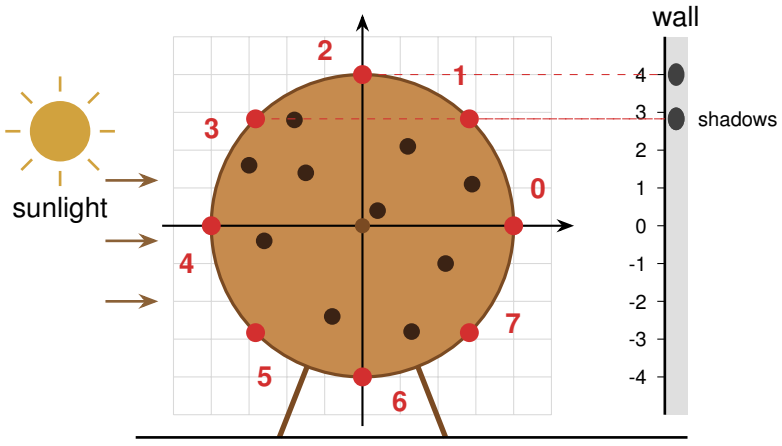
Remember Cookie Monster's giant cookie from The Cookie Circle? Now he has stood it up on its edge and made it into a **Cookie Ferris Wheel**! It turns around and around at the same speed. **Elmo** rides in a seat on the edge.

The sun shines in from the side, so Elmo's **shadow** lands on a wall. As the wheel turns, the shadow **slides up and down** the wall. It never gets bigger or smaller: it is always level with Elmo. **The Count** points: *"Elmo goes around and around. But his shadow goes up and down! Ah ah ah!"*

That up-and-down shadow is a **wave**. Radios, phones, Wi-Fi, light, and sound are all made of waves like this. Let's draw one.

Elmo's shadow on the wall

Here is the wheel from the front, drawn on a grid. Its edge is **4 squares** from the middle. Elmo's seat starts at spot **0**, level with the middle, and the wheel turns so he goes **up** first. Each spot is **one eighth** of a lap. We measure Elmo's **height** from the middle of the wheel: above the middle is plus, below is minus. The dashed lines show that his shadow on the wall is always at the same height as Elmo.



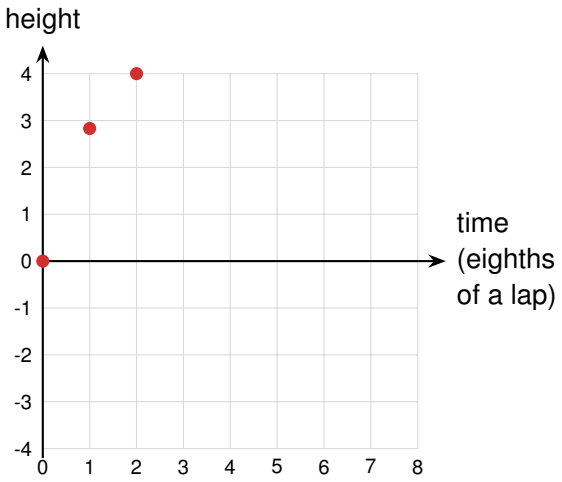
Question 1. Fill in Elmo's height at each spot. It is also where his shadow is on the wall. (Spots 1, 3, 5 and 7 are a little less than 3 squares, so write "about 3.")

Spot	0	1	2	3	4	5	6	7	8
Height	0	3	4						

Spot 8 is back at the start: one whole lap!

Draw the wave

Question 2. Put a **red** dot for each height from your table: spot number across, height up. Then connect the dots with a smooth, curvy line.



Going **around** a circle makes a **wave** that goes **up and down**! This wave has a name: the **sine wave** (say "sign"). Every lap of the wheel draws one more bump up and one more dip down.

Three words engineers use

Word	What it means for Elmo
Amplitude	How **tall** the wave is. It is the size of the wheel: how far the edge is from the middle.
Frequency	How many **laps** the wheel turns every second. The Count counts them: *"ONE lap! TWO laps! THREE laps in one second! Ah ah ah!"* That is 3 **hertz**.
Phase	**Where on the wheel** you are, or how far **ahead** one rider is of another. We measure it in turns, or in degrees: a whole turn is $360°$.

Question 3. The wave's highest point is _____ squares. So its amplitude is _____.

Question 4. A radio station at 100 megahertz makes its wave go around **100 million** laps every second! If Elmo's wheel turns 5 laps in one second, his frequency is _____ hertz.

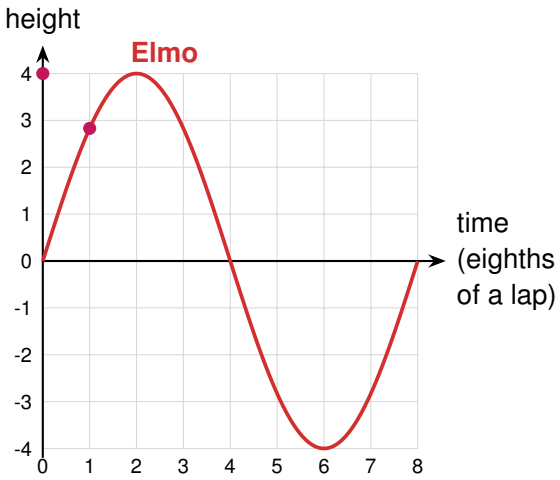
Phase: Abby starts ahead

Abby Cadabby rides too. Her seat starts at spot **2**, at the **top** of the wheel, a **quarter turn ahead** of Elmo. They go around together on the same wheel.

Question 5. Fill in Abby's height when Elmo is at each spot. (Abby is always 2 spots ahead.)

Elmo's spot	0	1	2	3	4	5	6	7	8
Abby's height	4	3							

Question 6. Draw Abby's wave in **pink** on the grid. Elmo's wave is already there.



Question 7. Abby's wave looks just like Elmo's, but **slid over**. Whose wave reaches the top first?

_____

Abby's wave is the same wave, **shifted**. That shift is called the **phase**. Abby is a quarter turn ahead, so her phase is $90°$ ahead of Elmo's. Engineers say the two waves are $90°$ **out of phase**.

Question 8. Fill in the phase for each turn.

How far ahead	Phase
quarter turn	$90°$
half turn	
three-quarters turn	
whole turn	

Oscar's opposite wave

Oscar the Grouch rides in the seat at spot **4**, a **half turn** $(180°)$ ahead. When Elmo goes up, Oscar goes down. *"Scram!"*

Question 9. Fill in Oscar's heights, then **add** Elmo's and Oscar's heights together.

Spot	0	1	2	3	4	5	6	7	8
Elmo	0	3	4	3	0	-3	-4	-3	0
Oscar	0	-3							
Elmo + Oscar	0	0							

Question 10. What happens when you add two waves that are a half turn apart?

_____

Waves a half turn apart **cancel**: together they make **nothing**! Waves that start together (**in phase**) **add up** to a bigger wave. This is called **interference**.

Question 11. Elmo's twin rides in the same seat as Elmo, in phase. Add their heights at spot 2: $4 + 4 =$ _____. The new wave is _____ (taller or shorter?).

How engineers use phase

- **Noise-canceling headphones** listen to the noise and play the same wave a half turn ahead, so the noise cancels.
- **Antennas** working together add their waves in phase in one direction, so the radio hears farther, and cancel in other directions.
- **Radios** send messages by changing the phase of a wave: a little turn ahead or behind means a different message.

Back to Euler's equation

The Count's Math Club: Euler's wave equation

In The Cookie Circle, $e^{j \times \text{steps}}$ meant "go that many steps around the circle." Engineers write the steps as θ (the Greek letter "theta"). Euler showed:

$$e^{j\theta} = \underbrace{\cos\theta}_{\text{floor shadow}} + j \underbrace{\sin\theta}_{\text{wall shadow}}$$

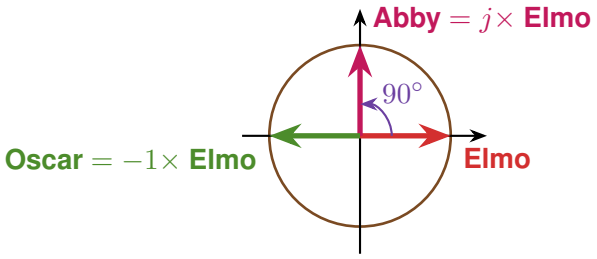
The **wall shadow** (sun from the side) goes up and down: it is the sine wave you drew. When the sun is straight **overhead**, Elmo's shadow lands on the **floor** and slides side to side. That is the **cosine wave**, and it is the same as Abby's wave! Radio engineers call these two shadows **I** and **Q**.

The big secret: a phase shift is a turn you multiply by. Abby is a quarter turn ahead of Elmo. A quarter turn is j! So

$$\text{Abby} = j \times \text{Elmo}.$$

Oscar is a half turn ahead: Oscar $= -1 \times$ Elmo. That is why his wave is upside down, and why Elmo $+$ Oscar $= 0$.

Question 12. Which number do you multiply by to shift a wave a quarter turn? _____ A half turn? _____



Engineers draw each wave as an **arrow** on the wheel, called a **phasor**. The arrow's **length** is the amplitude. The arrow's **direction** is the phase. As time goes by, all the arrows spin around together, like Elmo, Abby, and Oscar riding the wheel. One arrow tells you everything about a wave!

Draw it!

Draw a cookie wheel with three arrows for Elmo, Abby, and Oscar. Next to it, draw their three waves.

Answer key

For teachers: the wheel is vertical and seen face-on, and sunlight makes the rays parallel, so each shadow is an exact projection. Elmo's position after angle $\theta = \omega t$ is $4e^{j\omega t}$; his wall shadow is $4\sin\omega t$ (the quadrature, Q, component) and the floor shadow is $4\cos\omega t$ (in-phase, I). The heights at the eighth-turn spots are $4\sin 45^\circ \approx 2.83$, written "about 3." Abby's wave $4\sin(\omega t + 90^\circ) = 4\cos\omega t$ equals the imaginary part of $j \cdot 4e^{j\omega t}$: a phase shift ϕ is multiplication of the phasor by $e^{j\phi}$. Oscar's $\phi = 180^\circ$ gives $e^{j\pi} = -1$ and destructive interference. The radio examples are noise cancellation, phased arrays, and phase-shift keying. Sesame Street characters appear by name only, for classroom use; this worksheet is not affiliated with or endorsed by Sesame Workshop.

Question	Answer
1	0, 3, 4, 3, 0, -3, -4, -3, 0 (the 3s are about 2.83)
2	One smooth bump up (spots 0 to 4) and one dip down (spots 4 to 8)
3	4; 4
4	5 hertz
5	4, 3, 0, -3, -4, -3, 0, 3, 4
6	Same shape as Elmo's, starting at the top
7	Abby's
8	Half turn: 180°; three-quarters turn: 270°; whole turn: 360°
9	Oscar: 0, -3, -4, -3, 0, 3, 4, 3, 0. Elmo $+$ Oscar: 0 at every spot
10	They cancel: the sum is always 0
11	8; taller (twice as tall)
12	j; -1